

AW

NAS INTELLIGENCE PLATFORM

AI Work Portal

A NAS-centric workspace for enterprise multimodal AI

Unify NAS intake, meeting recording, speech transcription, image and document OCR, Hybrid RAG, generative models, LINE collaboration, and complete auditability. Every step from source intake to AI answer remains traceable and manageable.

24x7

NAS intake and live status

8+

Core AI workflows

2-tier

Exact and semantic cache

ZH / EN

Bilingual interface

All persistent information stays on the NAS

Source assets, transcripts, translations, OCR text, page images, RAG chunks, vectors, answer cache, model weights, and call audits remain on NAS storage. Only the content required for an explicitly selected cloud operation is sent to that provider.

NAS data sovereignty

Source and derived AI content share one asset lineage for access control, versioning, reprocessing, and snapshots.

Local and cloud choice

Select NAS-local models or company-governed cloud APIs per workload.

Discovery and automation

New-file notifications route content to ASR, OCR, vision, or RAG by type.

Enterprise governance

Centrally manage accounts, LINE groups, model quotas, data scope, and complete call records.

AI Work Portal | NAS-first AI, governed for enterprise use

A NAS-first data lifecycle

Each file receives a NAS asset ID linking its source, owner, processing stages, model output, RAG index, and audit trail. AI results never overwrite source files.

Secure intake → Live discovery → Parse → NAS index → Retrieve / answer

Meetings and audio

Browser recordings and uploads preserve source voice before local or cloud ASR. Source transcripts, translations, and model provenance are indexed separately.

PDF, DOCX, and images

Extract text PDFs directly; process image PDFs and JPG/PNG with PyMuPDF/PaddleOCR while retaining page, source preview, and chunk type.

Video analysis

YOLO samples frames and stores objects, confidence, timestamps, and previews as searchable video_detection chunks.

Cloud analysis persistence

Local files may be analyzed by cloud models; answers, model metadata, and citations are automatically saved to NAS and can feed later RAG.

What is retained on the NAS

Class	Retained content
Source assets	Audio, video, images, PDF, DOCX, and general files with filename, MIME, size, uploader, and timestamp.
Derived content	Transcripts, translations, OCR, video detections, summaries, LLM answers, and source previews.
Knowledge index	Chunks, pages, chunk type, 1024-dimensional embeddings, metadata, and access scope.
Operations	Processing state, errors, model I/O, caller, tokens, balance, cache hits, and timestamps.

Live status

WebSocket notifications cover intake, processing, completion, missing-model, and failure states. Each asset shows its parser, model, and processing timeline.

Models, Hybrid RAG, and enterprise governance

Local generation and vectors

Qwen3-4B-Q4_K_M runs through llama.cpp with an OpenAI-compatible API; Qwen3-Embedding-0.6B creates 1024-dimensional vectors on CPU.

Speech models

Choose whisper.cpp, faster-whisper, SenseVoiceSmall, or Paraformer-zh; cloud options include OpenAI Transcribe, Deepgram, and AssemblyAI.

Cloud LLMs

The provider catalog includes OpenAI, Claude, Gemini, DeepSeek, Qwen, Mistral, Groq, and more, with pricing snapshots, key requirements, and free-tier context.

Model management

Discover models and runtime readiness; show size, path, integrity, and progress with add, download, cancel, retry, and refresh controls.

Hybrid RAG and answer caching

75%

Vector similarity

25%

Keyword score

0.76

Semantic cache threshold

2-tier

Exact + semantic

Before invoking an LLM, the portal checks exact and semantically similar NAS answers, then runs hybrid vector and keyword retrieval. Cache entries are isolated by asset and model and invalidated when content changes, reducing duplicate cloud API calls.

Account management

Administrators create accounts, assign roles, disable access, reset passwords, and revoke active sessions.

LINE enterprise management

Approve groups, select company models, enable PDF summaries and @Claire RAG, and set monthly call/token quotas. Direct messages cannot use company resources.

Complete model audit

Successes, failures, and cache hits retain time, caller, provider, model, input, output/error, token data, and access mode.

Services and isolation

FastAPI, generation, embedding, ASR, and vision manage dependencies and devices independently and can run under systemd restart policies.

Layer	Technical baseline
Web / API	FastAPI, session cookies, role-based access, Traditional Chinese / English UI
Persistence	NAS filesystem + SQLite for assets, meetings, chunks, vectors, cache, and call audits
Document / vision	pypdf, PyMuPDF, PaddleOCR, python-docx, and YOLO
Integration	OpenAI-compatible APIs, mainstream cloud models, and LINE Messaging API

Enterprise value

Keep sensitive data and vectors on the NAS, use local retrieval and cache first, and invoke local or cloud models only when needed to balance control, cost, quality, and traceability.